Halo X Edge Impulse: Pet Activity Proposal
2.18.26

What we know

This isn't a simple fix - and that's exactly why we're here.

- We're stepping into an existing system, which means before anything else, we need to fully understand what's been built. From there, we define the path forward together.

What this will take

- Time and iteration - the right solution won't come from a quick patch
- A willingness to redesign - the current system architecture may need to be rethought
- Full access to your data, system, and team
- Close collaboration from both sides, at every stage

Halo Collar - Edge Impulse Proposal

The Edge Impulse Approach:

- **Innovation & Iteration —** we define the solution through testing, refinement, and close collaboration
- **Built on EI's Full Capabilities** — specialized testing ensures identical preprocessing and model processing from cloud to device, eliminating surprises at deployment
- **Full Visibility —** your team gets full access to Edge Impulse, a single transparent environment where every model, dataset, and decision lives
- **We consult and enable your engineers —** your team owns the solution long-term, not dependent on us to keep things running
- **Proven in the pet space —** including activity detection work with Nestlé Purina

What's Included:

- Edge Impulse platform access · Dedicated advanced engineering support

Total Annual Investment

- $133,000 — Edge Impulse platform + advanced technical support
- $8,000–$30,000 — One-time commercial deployment license, scaled to device volume

How We Work Together

- **Regular syncs to align on priorities and surface blockers early** — more frequent during critical phases
- **Data collection strategy** — guidance on sensor placement, sampling rates, and data diversity to build a robust training set
- **Model development & iteration** — training and refining activity detection models tailored to wearable collar sensor data
- **Validation & testing** — rigorous evaluation against real-world conditions, not just held-out test sets
- **Edge deployment & optimization** — ensuring models run efficiently within the power and memory constraints of the collar hardware
- **Documentation & knowledge transfer** — your team understands every decision made, not just the final output
- **Ongoing performance monitoring** — tracking model performance overtime with continued model iteration and improvement

Phased Approach

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Discovery & Alignment	Experimentation & Validation	Optimization & Commercialization
• Assess your current system end-to-end • Explore your data inside Edge Impulse • Align on gaps, opportunities, and a clear path forward • Review with ML and embedded teams at Edge Impulse	• Restrict Scope and focus on the highest-value activity classes first • Validate modeling and preprocessing assumptions early • Rapid pipeline exploration using EON Tuner	• Test performance against real-world data • Integrate seamlessly into your existing workflow • Optimize and refine model to prepare for commercial deployment

Premium Support Workstreams

Proposed scope — subject to refinement following discovery and data review.

Workstream	Objective	Key Tasks	Level of Effort
1: Windowing & Inference Pipeline Review	Align windowing with real behavior	Map flow; validate; document limits	1-2 weeks
2: Orientation Assumption Validation	Quantify collar orientation impact	Field study; sensitivity analysis	2-3 weeks
3: Class Simplification & Data Strategy	Focus on core behaviors	Redefine classes; update labels; collect data	3-6 weeks
4: Evaluation Framework Redesign	Use real-world metrics	Define KPIs; build harness; confusion matrix	2-3 weeks
5: Application-Level Logic	Stabilize predictions	Smoothing; gating; state machine	2-4 weeks
6: Model & Feature Exploration	Reduce bias; test architectures	New features; CNN/Transformer tests	4-8 weeks

Notional High-level Timeline

Proposed scope — subject to refinement following discovery and data review.

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Phase	Duration	Workstreams
1: Analysis and Foundational Review	Weeks 1-3	WS1, WS2
2: Data Strategy and Class Simplification	Weeks 3-8	WS3
3: Modeling & Evaluation Redesign	Weeks 5-12	WS4, WS6
4: Application Logic & Integration	Weeks 8-16	WS5
5: Final Recommendations	Weeks 14-16	All work streams converge